Prototyping Plan

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	04-09-2026	Draft Version	Casey Stijlaart	Draft Version	Split out from the Technical Research Report's non-camera-sensing-techniques chapter into its own document
0.2	08-09-2026	Draft Version	Casey Stijlaart	Draft Version	Revision based upon feedback from Gerard Elbers.

This report carries the mmWave radar recommendation from the Technical Research Report's Conclusion into a concrete plan for the bench-experimentation stage, built up in capability-based phases rather than deferred behind a cheaper sensor.

2. Approach

Build the mmWave pipeline once, adding one capability at a time on top of an already-working previous phase, rather than assuming position tracking, height estimation, multi-person clustering, and speaker detection all work together on the first attempt.

2.1. Phase 1: Position Tracking

Bring up the core sensor-to-output pipeline before any height, multi-person, or speaker-detection logic is added:

- mmWave sensor driver: UART point-cloud parser for the IWR6843AOPEVM.
- Clutter removal, clustering, and Kalman filtering for position tracking, following Huang et al. (2021).
- An Unreal Engine socket bridge (C++ TCP output).

This phase validates the parts of the system every later phase depends on: sensor bring-up, point-cloud parsing, and the Unreal integration. A failure here is a failure in the foundation, so it is deliberately tackled before any of the more research-heavy capabilities below.

2.2. Phase 2: Height Estimation

Extend the working (x, y) position tuple from Phase 1 to (x, y, z):

- Choose between a lightweight blob-size heuristic and a keypoint-extraction model following ProbRadarM3F-style approaches (2024) and Zheng et al. (2025), trading implementation effort against accuracy.
- Feed the resulting height value into the same output tuple and Unreal bridge already validated in Phase 1.

Fallback: if mmWave's own height estimate underperforms once measured, the mmWave + Thermal Fusion for Height option documented in the Technical Research Report's Candidate Sensor Technologies chapter is the next step, adding a thermal array as a second, height-specific sensor rather than replacing mmWave.

2.3. Phase 3: Multi-Person Clustering

Extend the Phase 1 clustering step to reliably separate multiple simultaneous people in the point cloud, rather than assuming a single occupant, following Huang et al.'s (2021) multi-person tracking demonstration. This validates RQ4's extensibility requirement at the architecture level before speaker detection is added on top of it.

2.4. Phase 4: Speaker Detection

Add mmWave vibrometry to localise and steer toward an active speaker's throat, reusing the same radar hardware and multi-person clustering from Phase 3, following Xu et al.'s WaveEar (2019). This completes RQ4's active-speaker identification requirement using the same sensor family throughout, rather than a second, dedicated device.

Risk mitigation: because each phase adds a capability on top of an already-working previous phase rather than swapping hardware, a problem in one phase is isolated to that phase; if Phase 4's speaker detection proves difficult, Phases 1-3 remain a working position/height/multi-person system rather than a dead end.

NOTE	Phase durations are not fixed here; they should be mapped onto the sprint structure in Timeline, since that document is the authoritative one for actual dates.

3. Next Steps

- Confirm the physical dimensions of the Holobox interaction space, since this determines the effective range required from the mmWave sensor.
- Present this plan and the mmWave hardware cost estimate to the company mentor for budget approval, per the process in Finance and Risks; no hardware is ordered before that approval.
- Once approved, procure the mmWave module and begin bench experimentation with Phase 1, as described earlier.
- Treat Phase 4's speaker detection as a follow-on capability to validate once Phases 1-3 are working, rather than a day-one requirement.

4. Bibliography

Huang, X., Cheena, H., Thomas, A., & Tsoi, J. K. P. (2021). *Indoor Detection and Tracking of People Using mmWave Radar*. Journal of Sensors, 2021, Article 6657709. https://onlinelibrary.wiley.com/doi/10.1155/2021/6657709

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